

forward from the current portfolio and buy and sell shares efficiently until we arrive at the optimal end point. Proceeding in this manner will ensure that we only take an efficient implementation path or only a difficult path if the end return more than offsets the incremental cost. The end product in this case is determined directly from the trade schedule and ensures the most frictionless path was taken.

Quest for best execution frontier

The quest for the best execution frontier is centered on proper integration of trading costs into the portfolio optimization. The trick is to incorporate a variable market impact function dependent upon the number of shares transacted (size), volatility, trade strategy, and the overall risk composition of the trade list (covariance) to take advantage of potential diversification and hedging opportunities.

For consistency of notation and uniformity across trading horizon and holding periods we express decisions in terms of dollars and shares to trade rather than the traditional investment units of weights and returns. This is also more important because it is the dollar value and shares traded that affect market impact cost and not the weight of the stock in our portfolio.

Using these new units, the original portfolio construction optimization problem (Equation 10.3) can be written in terms of a cash investment and shares to trade as follows:

$$\begin{aligned} \underset{S}{\text{Max}} \quad & S'(P_t - P_0) + \lambda \cdot S'CS \\ \text{s.t.} \quad & S'P_0 = V_S \end{aligned} \tag{10.8}$$

where S is the vector of shares to hold in the portfolio (decision variable), P_t is the vector of expected prices at time t , P_0 is the vector of current prices, and C is the covariance matrix expressed in $(\$/\text{shares})^2$.

To properly incorporate trading costs into Equation 10.8 it is necessary to introduce a new decision variable x_k to denote how shares are to be transacted over time (e.g., the underlying trading strategy). This solution is best accomplished via a multi-period optimization formulation that considers both a trading horizon where investors acquire shares from $t = 1$ to $t = n$, and an investment or holding horizon where no other shares are transacted from $t = n$ to $t = T$.

Let us now consider the effect of trading costs on portfolio return and risk in terms of the multi-period context.